Honours Project - MHW225671

FINAL REPORT

2025-26

Department of Computer Science

Submitted for the Degree of:

BSc (Hons) Computing

Project Title: Usage tracking & psychology-based mobile application to help address social media and smartphone addiction in young adults

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Second Marker: To be confirmed

Word Count: 10000

(excluding contents pages, figures, tables, references and appendices)

Aligned to SDGs: 3, 4

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Abstract

Throughout the rise of social media and smartphones, an ever-growing topic of discussion has been the issue of individuals developing an overdependence on said technologies, referred to as "Social Media Addiction" and "Smartphone Addiction" respectively. Research into the topic has been performed extensively in the past, with some looking towards psychological-based treatments such as Cognitive-Based Therapy and Reality Therapy, and in recent years, some looking at mobile applications that use some form of usage tracking to address the issue. Research has shown that psychological-based treatments has provided promising results in treating similar issues such as anxiety, and has even seen integration with mobile apps. Similarly, studies conducted show viability in utilising usage tracking when addressing the issue, particularly when integrated into a mobile app. However, little work has been done to utilise both of these concepts together into a single mobile app.

This project showcases an Android-based mobile application that integrates concepts from Reality Therapy alongside usage tracking technologies, with the overall objective being to lower an individuals usage in an attempt to address the issue of Social Media and Smartphone Addiction. To evaluate the project, a small group of participants were given the application to use over a three-week period, and were asked to send their usage data each week during this time. By the end of this period, they were also provided a survey where they could provide their own opinions on the app.

Results from the evaluation show that the PRI was somewhat successful in reducing the usage of smartphones and social media in young adults by using concepts from Reality Therapy and usage tracking technologies.

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Abbreviation / Acronym	**Description / Definition**
RT	Reality Therapy
WDEP	Wants, Direction, Evaluation, and Planning & Commitment
SRS	Software Requirements Specification
PRI	Primary Research Instrument
SDLC	Software Development Lifecycle
SA	Smartphone Addiction
SMA	Social Media Addiction
LSEP	Legal, Social, Ethical, and Professional
QS	Question Set

Table 1: List of Abbreviations and Acronyms

1 Introduction

This section will provide context on the issue of smartphone addiction and social media addiction, including how people develop such addictions, and the effects they have on an individual. Existing apps and studies that attempt to deal with the issue are discussed. It will also provide an overview on the project and it's objectives to be achieved in the literature & technology review and during development.

1.1 Project Background

Social media (SM) and smartphones have been adopted as an important piece of daily life for a substantial number of people in the world. They have provided us the ability to connect with from around the globe on a never-before-seen scale, allowing for friendships, communities, and relationships to be built from virtually anywhere. Dating back MySpace in the mid-2000s, the release of Apple's "iPhone" and the release of the HTC Dream running the "Android" operating system in 2008, to SM such as Twitter and Facebook reaching out to billions around the globe, smartphones and social media looks to be omnipresent within a majority of peoples' lives, with it being estimated that 1 in 3 people globally have some form of SM presence, reportedly 85% of adults in the United States own a smartphone .

Over the years, worries have been raised regarding many issues that concern smartphones and SM, particularly surrounding the problem of individuals developing an overdependence on their smartphones and SM that negatively affects their day-to-day life . Commonly categorised as "Smartphone Addiction" (SA) and "Social Media Addiction" (SMA) and considered subsets of behavioural addiction , the issue has been largely discussed and researched over the last 15 years, with researchers looking into potential causes and correlations with other health effects.

analyze SA and provided potential causes behind SA, one of which being how the current generation has been brought up alongside the internet and technology, therefore possibly feeling a preference to communicate over the internet instead of in-person, fuelling the dwindling need for face-to-face communication. made a series of hypotheses in relation to college students and the effects related to SA, looking at SA's associations with "life stressors" (a measure of stress and anxiety), "digital overload" (a measure of information and communication overload), retention, academic performance, and social well-being. The results found that SA could be positively correlated with life stressors, digital overload, retention, and academic performance, something consistent with previous studies.

Potential treatments have been researched as well. A China-based study from

2019 analysing SMA developed a cognitive-behavioural-therapy-based intervention program that used various techniques including reminder cards, journal entries before sleeping, and cognitive reconstruction. The results showed that compared to the control group, those who took part in the program saw improvements in mental health, engagement in learning, sleep quality, and also saw a drop in SM usage.

The "AntiSocial" app, developed by Australian company BugBean for Android devices allowed users to restrict access to certain apps and provided app usage statistics, but also added other functionality like usage comparisons with others that were within their age group, location, and occupation. It also showed users how many times they picked up their device, and how long they used certain apps for . The author stated that the app ranked 3rd in downloads on the UK Google Play store mere days after launch, "demonstrat[ing] the need for [AntiSocial.io]"

However, there seemingly exists a gap of intervention apps that deal specifically with SA or SMA, and that also allow self-reflection from the user, and most literature on the topic mostly provide analysis on apps that deal with adjacent issues. A 2021 study by reviewed the effectiveness of psychological intervention mobile apps, and while they concluded that the apps showed promise, none of the literature specifically looked at addressing the issue of SA or SMA.

1.2 Project Overview, Aims, and Methodology

1.2.1 Project Outline

Evidence has shown that smartphone apps that help with SA and SMA are in-demand, and significant amounts research have been conducted suggesting treatments to the issue. However, most consumer products on the app store opt to utilize app-restriction features. Because of this approach, young adults who use these apps aren't given much opportunity to reflect on the nature of SA & SMA and the potential causes of it, leading to very little character growth for the end-user. However, by taking a different approach by integrating psychology, young adults can be given the opportunity to reflect on their addiction and learn how to address the issue. With that being said, the research statement is as follows:

To create a mobile application that integrates psychological intervention methods alongside usage tracking technologies to help address smartphone and social media addiction

1.2.2 Project Objectives

The primary objective of this project is to create an application that can guide the user into reducing their smartphone & social media usage with the use of usage tracking and psychological methods. This type of app can be considered a kind of "persuasive computing", a type of computer system intended on changing the behaviours and attitudes of a user .

Secondary Objectives

Investigate existing mobile apps that use psychological methods to address mental health issues
Research into how existing mobile applications make use of psychological methods when addressing issues such as depression and anxiety to understand how viable using mobile apps are to address similar issues.

Investigate the contextualization of SA and SMA in existing apps
Research how the issue of SA and SMA are contextualized within existing intervention apps and what issues could arise from how they are presented. This will help in enabling growth in a user when it comes to learning about SA and SMA.

Investigate psychological frameworks
Research how psychological methods have been used in the past, specifically to treat SA or SMA, and see if such methods have been integrated with mobile apps to gain a further understanding on how these methods could be integrated or improved upon.

Investigate methods to track app usage
Research into how apps currently track phone activity, and how they can be utilized in the context of the project, as a major component of the application makes use of usage tracking.

Primary Objectives

Develop the smartphone intervention app
Create the application based on the knowledge gained from the literature and technology review. This will involve integrating the chosen psychological framework and usage tracking method before testing the app's functionality via test cases to ensure the app functions as intended.

Gather participants for evaluating the app and perform evaluation
Participants will be used when evaluating the developed application. They will be

given an overview of the app before being given a consent form to sign to consent to participating in the research.

Analyze the results and obtain a conclusion
Collect results from the evaluation and see if the app was impactful at reducing smartphone and social media usage. Then evaluate based on the results, and see if the problem statement and hypothesis were met.

1.2.3 Project Hypothesis

Testing the hypothesis will involve providing participants with the developed application for a period of three weeks. During this period, participants will attempt to reduce their smartphone & social media usage by using the developed mobile application with usage tracking and psychological intervention methods. Given this, the hypothesis for this project is:

The combination of usage tracking and intervention methods rooted in psychology within the created mobile application will help to address and reduce smartphone & social media usage in young adults.

1.3 Structure of the Report

Before continuing with the remainder of this paper, a brief overview of the structure of the report is provided, giving a short summary of each upcoming chapter.

Chapter 2 - Literature and Technology Review - This chapter provides an extended review of the current literature surrounding topic areas relevant to the report while also analysing technical methods utilised by apps with a similar intention to the PRI outlined within the primary objectives.

Chapter 3 - Execution - This chapter provides an extensive look towards the development of the PRI, providing commentary towards the research methodology used, and discussion and justification on the design choices, technological components used, features implemented when developing the PRI, and the results from testing the application for bugs.

Chapter 4 - Evaluation and Results Discussion - This chapter presents a in-depth discussion of the results of the evaluation, including the findings themselves, reasoning on the findings, and initial conclusions from the findings.

Chapter 5 - Legal, Social, Ethical, and Professional Issues - This chapter gives commentary on any Legal, Social, Ethical, and Professional the project encountered over its lifespan, with additional discussion provided towards the Sustainable Development Goals the project aligns with when discussing social issues.

Chapter 6 - Conclusions and Further Work - This chapter concludes the report by summarizing the project in it's entirety up to this point, before then presenting a final answer to the project's hypothesis. It then continues by providing critical analysis of the project and it's limitations, and finally, discusses what further work the project may lead in the future.

2 Literature and Technology Review

This literature and technology review will review mobile apps that use psychological methods to address mental health issues before looking at how existing apps provide context regarding SA & SMA and the issues associated with it. Potential psychological frameworks for treating SA & SMA are reviewed before researching how app usage can be tracked.

2.1 Investigate existing mobile apps that use psychological methods to address mental health issues

The project background mentioned that although researchers have studied SA and SMA, few apps are available, or have been created for research purposes that handle the issue. However, apps that address similar issues exist and have extensively been reviewed and evaluated. This section reviews existing studies on these apps to gain a further understanding on the viability of treating mental health issues.

2.1.1 Prior research on mobile app interventions

Researchers reviewed mobile apps that integrated psychological frameworks to treat mental health issues such as depression, anxiety, drug use, among others, with their intention being to evaluate the effectiveness of these apps in reducing mental health symptoms. The review discusses features used, with the most common being behaviour monitoring, participant engagement via explicit input, tailoring based on feedback, and gamification, while also noting the majority of apps opted to use CBT as their psychological framework. They discover that out of all the mental health issues targeted, depression, sleep-related issues, and smoking returned a significant effect, with the review stating that "our knowledge on how to design effective mental health apps is very much at the beginning".

They continue by suggesting different approaches that focus more on the user and the framework, as well as engagement with the app via usage patterns, before saying that at the current point in time, there isn't much significant evidence that supports using mobile apps for treating mental health as an alternative to traditional treatment, highlighting that while apps targeting depression, sleep issues and smoking showed promising results, the results showed a high risk of bias within them.

reviewed mobile app-based interventions utilizing psychological frameworks targeting college students, aiming to review the outcomes of using app-based psychological interventions instead of traditional therapy, as well as looking how these apps are being embraced by students. The review details the types of mental health issues

these apps target, while also noting that a majority of these apps are "self-guided" in nature . Psychological frameworks used are mentioned, with Cognitive Behavioural Therapy (CBT) being the main framework used or combined with another framework, with other frameworks such as Acceptance and Commitment Therapy and Self-determination Theory mentioned.

The review then concludes by saying that the current literature demonstrates that using mobile apps for mental health intervention is a viable option and could be a great resource for counselling services, noting that future work into the topic should utilize alternative psychological frameworks, look further into what students and therapists would want out of similar apps, and that future studies should stray from monetary incentives for students looking to participate .

2.1.2 Discussion

found that while apps that targeted depression, sleep-related issues, and smoking showed some results, the field was merely in its infancy when it came to designing effective applications that integrate psychological frameworks. Later, showed that mobile apps that targets mental health issues aimed towards college-level students were a viable alternative option for treating mental health issues due to student embracement, showcasing growth within the topic between the two points in time, though noted that this could've been due to incentives attached within the latter study.

Despite lacking insight on psychological frameworks other than CBT, showing a need to expand the types of psychological frameworks used for these apps, and the use of incentives which may bias user retention, the studies demonstrate that the viability of a mobile mental health app using psychological frameworks has grown, and that developing such app is doable, as long as aspects such as other psychological frameworks, and user incentive are considered. The studies also highlight that while apps such as the ones discussed are in their infancy and may not deliver the same results as traditional treatment, the ease of access compared to traditional treatment provides another point to its potential viability.

2.2 Investigate the contextualization of SA and SMA in existing apps

The project background mentioned that intervention apps are developed with mostly functionality in mind, while neglecting any sort of teaching element, therefore leaving the user little to no room to learn about SA and SMA. This section reviews how SA and SMA are contextualized, and how this could be improved.

2.2.1 Issues with contextualization

puts forward an argument stating that SA apps created by developers aren't suitable for treating SA due to developers having limited knowledge on the topic of SA, and points out the irony in attempting to treat SA with the use of a smartphone. This sentiment is shared in , stating that proposing technology as a solution to an issue caused by technology itself is paradoxical in nature.

Dosselaar says that developers lack a deep understanding of the topic, and points out the overall complexity of the issue, mentioning how SA is a product of many other factors; not just the phone, and the apps simplify what is often a complex issue by acting as a one-size-fits-all solution . They also fail to consider context behind smartphone usage, noting how it can be productive and non-disruptive depending on where it's used . Another issue is that the apps lack any kind of opportunity for the user to understand and reflect on the issue of SA, noting that these apps frame SA as an issue that can be quantified, while lacking any features that offer information about SA , something that plays a role in treating the issue , and could influence app effectiveness .

The author concludes by stating that SA apps are limited in effectiveness as they focus on practical functionality of the app rather than the contextualization and opportunity for reflection for the user, before recommending researching integrating psychological frameworks into the apps .

2.2.2 Discussion

While it can be argued that some of these issues have been addressed, for consumers, most of these issues are yet to be addressed by anyone but the developers of these apps . These apps focus only on features such as self-monitoring and app restriction , rather than providing opportunity for end-users to reflect and learn. These apps frame SA and SMA as an issue that can be solved with the apps themselves, rather than it being a complex issue , emphasised by the fact that not all smartphone usage is the same for each person , bringing suggestions of incorporating aspects such as psychological frameworks into these apps to make them more effective .

2.3 Investigate psychological frameworks

The project background mentioned that plenty of studies have been conducted utilising different psychological methods to treat SA and SMA. This section aims to review the most common psychological methods to understand which method would work best with the project.

2.3.1 Existing psychological frameworks for SA and SMA

The previous section highlighted the need for research into how psychological frameworks could be integrated into consumer SA and SMA intervention apps to make them more effective, a sentiment shared with , who mentioned future intervention apps should incorporate [other] psychological methods.

Intervention apps developed for research purposes commonly use Cognitive Behavioural Therapy (CBT) as their psychological framework , though other frameworks such as mindfulness, reality therapy, and theory of planned behaviour have also been utilized to mixed success .

2.3.2 Cognitive Behavioural Therapy

Cognitive Behavioural Therapy (CBT) is a psychological framework that take an "active, problem-focused, time-sensitive treatment approach" aiming to increase adaptive behaviour and reduce emotional distress in an individual with mental health issues via targeted, short-term strategies that intends to alter maladaptive emotional responses in an individual by changing their thoughts and/or behaviours . Despite results lacking in some areas, it can be argued that it's the framework with the highest usage overall, with some considering it the "gold standard" due to the proven success the framework has had on treating certain issues such as nicotine overdependence, schizophrenia, depression, and anxiety.

The main CBT techniques used in intervention apps are behavioural activation (A focus on engaging a patient in positive activities and helping with aversive control), cognitive restructuring (A process that assists patients in identifying and evaluating negative thoughts), and problem solving (Discover solutions to a patient's issue). assessed if these techniques could reduce smartphone usage via three separate apps which provided these techniques (AntiSocial, Headspace, Pacifica) and found that those in the experimental group saw their smartphone usage drop and were made more aware of their usage when compared to the control group.

2.3.3 Reality Therapy

Reality Therapy (RT) is a psychological framework that can be traced back to choice theory, which theorizes that all human behaviour is a result of choices, . RT's primary aim is on focusing an individual's behaviour in the present by using the Wants, Direction, Evaluation, and Planning & Commitment System (WDEP system) . This involves asking an individual to define a goal for themselves, what they're currently doing, and if that behaviour is halting progress towards their goal or not, before

being encouraged to replace it with a healthier alternative . RT has shown effectiveness in treating issues such as Internet Addiction and Internet Gaming Disorder , though has had limited research when treating SA and SMA compared to CBT.

developed a mobile app integrating RT. The app was hypothesized to reduce anxiety, SM usage, among others, and was developed for Android devices. It implemented teachings about SMA, as well as app restriction functions, with the results being that anxiety and SM usage fell when compared to the control group.

2.4 Investigate methods to track app usage

The project background mentioned that existing SA and SMA intervention applications mainly make use of app-restriction and usage tracking technologies, and due to usage tracking being a large component of the project, it would be useful to know the technological methods used for app usage tracking.

2.4.1 Justification for app usage tracking

App usage tracking has proven to be successful in reducing smartphone usage. a study by researchers reviewed the effectiveness of apps intended to reduce smartphone use, and found that in the 13 apps reviewed, 4 apps successfully reduced smartphone usage, those of which included app usage tracking.

2.4.2 Methods of app usage tracking

hypothesized that individuals with high levels of smartphone usage show more difficulty in not using their smartphone when compared to those with lower levels, verifying their hypothesis by developing an application called "SocialStatsApp" to track app usage of participants.

The app was developed for devices using Google's Android operating system. The researchers mention how making the app compatible for Apple's iOS devices wasn't possible as the OS disallows the collection of usage statistics for privacy reasons . The programming language used was Java, one of three officially supported languages via the Android SDK, the other two being Kotlin and C++ , though the study doesn't detail the reasoning behind using Java. The app makes use of the Android class
android.app.usage.UsageStatsManager from the Android API which gives access to phone usage stats and history. Because of this, the app needs permission manually granted by the user. The results of the study showed that those with high levels of smartphone usage did experience more difficulty in not using their phone.

analyzed the effects technology has on the mental health of young people, developing an app called the "3S-YP" app that constantly collected "smartphone metadata" from participants. While they don't detail the development of the app, they mention that smartphone metadata was collected using the android.app.usage package from the Android API, while also mentioning that metadata collection was not performed for the iOS version of the application due to "data protection restrictions", a sentiment shared by the previous study, showing that developing for Android allowed for additional functionality not possible with iOS.

Another method of tracking a user's app usage is by making use of screenshots and digital well-being features included with smartphones. investigates the use of digital health apps and their effectiveness via Apple's Screen Time feature. Due to the aforementioned privacy restrictions, the researcher instructs participants to instead screenshot multiple aspects related to the Screen Time feature. A limitation of this method however is that screenshots must be taken at a specified time, as mistiming it could yield less accurate metrics .

2.4.3 Discussion

Both studies chose Android as their development platform of choice, mentioning that iOS apps don't allow for app usage tracking due to privacy restrictions . This is elaborated further within , mentioning including iOS users would require a "jailbroken" phone, which brings a variety of issues. Meanwhile, Android allows for this as long as the user permits it. Only one of the studies makes mention of a programming language , that being Java, and while both studies make use of the android.app.usage package within the Android API, only one explicitly uses the UsageStatsManager class. Given this, it is clear that the OS that the app should be developed for is Android, and that using the android.app.usage package would be a viable method of app usage tracking.

3 Execution

This section will provide an in-depth look at the process behind building the PRI. This section will include a brief analysis on the research methodology used, and then a extensive look at the creation of the PRI, including commentary and justification behind design choices, technologies used, and the features implemented. When necessary, this section will also reference the literature review.

3.1 Research Methodology

The aim of this project is to create a mobile application that helps to address the issues of SA and SMA in young adults by integrating concepts and methodologies from RT alongside usage tracking technologies, then, with a group of users, test the application for a period of three weeks and assess whether the PRI meets the project objective and hypothesis.

The research methodology chosen for this project was a "develop and test" methodology. This methodology was appropriate for the objective as the primary objective of this project as stated in section 1.2.2 is to:

...create an application that can guide the user into reducing their smartphone & social media usage with the use of usage tracking and psychological methods.

and ensuring the objective is met involves testing the developed application with a small group of users, and evaluating based on captured data if the PRI meets the project's main objective, research statement, and hypothesis.

3.2 Lifecycle

Early in the planning stage of the project, the project intended to make use of the agile methodology, due to its focus on flexibility, adaptability to any changes in requirements, and frequent use in mobile app development. However, during the requirements analysis stage of the project, a change to a Hybrid methodology was made. This model aims to let development teams combine approaches from multiple methodologies and tailor the Software Development Lifecycle to fit their needs . In this case, approaches from iterative and waterfall methodologies were combined to create a methodology that fit the needs of the project at that point in time. specifically, the waterfall methodology was used for the Requirements Analysis, Design, and Evaluation stage, while the iterative methodology was used for the Implementation and Testing stage.

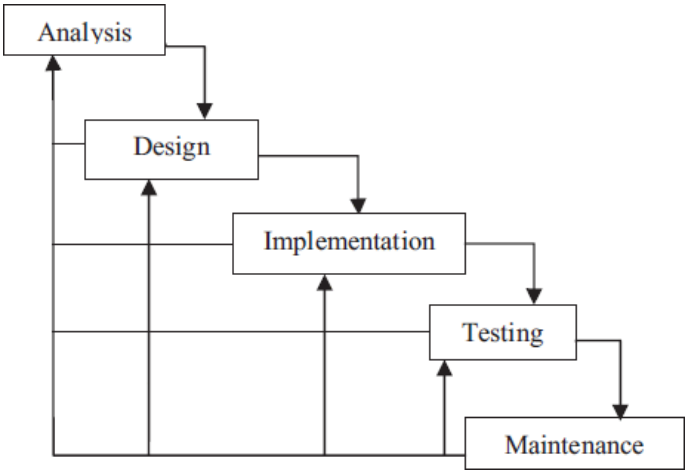

Figure 1: *Typical iterative Waterfall methodology*

This change in methodology was deemed to be necessary, as it felt more appropriate for the project as a whole when compared to agile and waterfall separately. Agile methodologies, while fitting for any software project, puts a focus on constant customer feedback during the SDLC , something that there would be a lack of in this project's case. The Waterfall methodology, while a lot simpler to plan with, its lack of flexibility when it comes to changes in project scope and lack of prototype creation in particular , made it usage unfitting for the project. Combining parts of these methodologies however, made for a methodology that was fitting for the project as a whole.

this change did not negatively affect development, despite this shift occurring slightly into the execution of the project,

The Hybrid model was used by using a waterfall approach for the Requirements Analysis, Design, and Evaluation stage, and combining it with an iterative approach for the Implementation and Testing stage. The requirements were defined during the beginning of the project's execution stage within a Software Design Specification (see Appendix F), then, based on the requirements specified, designs surrounding those requirements were created.

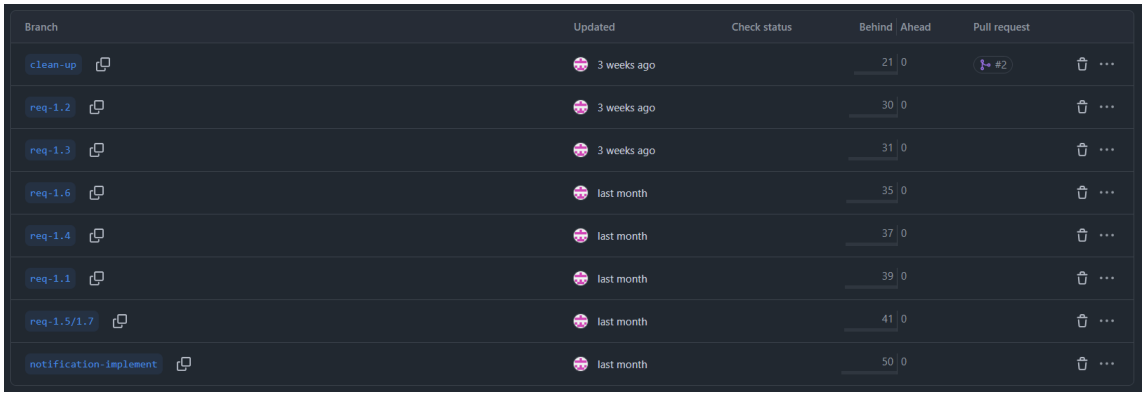

Figure 2: *GitHub branches for each requirement*

Figure 2 showcases the process behind implementing and testing the PRI. A GitHub branch that matched the name of the requirement was created, then, within that branch, that requirement was implemented, then tested upon via Incremental Integration Testing to ensure the requirement was implemented as intended. If that part of the requirement passed all test cases, a new GitHub branch for the next requirement was created using the newly completed requirement as a blueprint, and this process repeated until the implementation of all requirements were completed.

3.3 Requirements Analysis

When performing the requirements analysis for the project, the research statement as stated in section 1.2.1 was analysed to identify the main components of the application. Combining this with the knowledge gained after performing the literature review provided the blueprint from where functional (and non-functional) requirements could be identified. As a reminder, the research statement was:

To create a mobile application that integrates psychological intervention methods alongside usage tracking technologies to help address smartphone and social media addiction

From the research statement alone, two integral components can be identified:

1. A psychological intervention method that can be integrated into a mobile application
2. A code-based technical method of tracking smartphone and social media usage

From here, these components can then be broken down into smaller parts that then make up a list of functional requirements after ensuring they fit within the findings of the literature review, with some aspects from both components being used to identify a requirement. Additionally, interviews were conducted with nine participants to help identify new requirements or to add pieces to existing requirements. Full interview transcripts, instructions on accessing them, and detailed notes taken during each interview can be found within Appendices D and E.

3.3.1 Psychological Intervention Method

The psychological intervention method is a critical componentfor the PRI, as this would then ben used to derive functional requirements of the application, therefore, choosing a psychological intervention method came down to how easy it would be to integrate said method into a mobile app and how impactful the method itself is when used. The literature review, specifically section 2.3 made mention of two psychological intervention methods, Cognitive Behavioural Therapy and Reality Therapy.

After analysing both frameworks, it was concluded that RT was the choice of framework to use. The primary reason for using this framework instead of CBT ties back to one of the main criticisms of SA & SMA mobile apps found within the literature review, that being that current apps on the market do not provide any opportunity for end-users to be made aware of their usage or SA & SMA as

a whole. RT's framework inherently gives a person the opportunity to understand what's stopping them from achieving their goal, where in this case, their smartphone usage and/or SA & SMA would be the aspect affecting them. Meanwhile, CBT in its framework doesn't inherently provide this opportunity.

3.3.2 Usage Tracking Method

Finding a usage tracking method was a simple task. As mentioned in section 2.4, Apple's iOS operating system disallows apps from tracking a user's usage due to privacy concerns, with the only "viable" alternative being using a "jailbroken" device, which would complicate multiple parts of the project. On the contrary, Google's Android operating system allows for apps to track a user's usage via the "android.app.usage" package within its API, so long that a user permits the app to do so.

```
android.app.usage

@SystemService(Context.USAGE_STATS_SERVICE)
public final class UsageStatsManager

Provides access to device usage history and statistics. Usage data is aggregated into time intervals: days,
weeks, months, and years.

When requesting usage data since a particular time, the request might look something like this:

  PAST                    REQUEST_TIME                    TODAY                   FUTURE
  ---------------------------||---------------------------¦----------------------|
                          YEAR ||                          ¦                      |
  ---------------------------||---------------------------¦----------------------|
   MONTH            |         ||                MONTH      ¦                      |
  -------------------|---------||---------------------------¦----------------------|
    |      WEEK     |     WEEK||    |     WEEK     |     WE¦EK     |      WEEK     |
  ---------------------------||-------------------|-------¦----------------------|
                             ||           |DAY|DAY|DAY|DAY¦DAY|DAY|DAY|DAY|DAY|DAY|
  ---------------------------||---------------------------¦----------------------|
```

Figure 3: *UsageStatsManager class from android.app.usage, taken from Android's API documentation within Android Studio*

The "android.app.usage" package contains many classes that provide detailed statistics about a user's device, namely the "UsageStatsManager" class. Figure 3 shows how this class returns a request for usage data, with the documentation noting that requesting usage data within a time interval will include that interval. This class was mentioned within the literature review when discussing what methods could be used to track usage of a user's device, and is the class of choice for developing the

app, as it contains a method "queryUsageStats" that can be used to retrieve usage stats within a specified time range .

3.3.3 Functional (and Non-Functional) Requirements

As mentioned at the beginning of this section, requirements were identified by taking the findings from the literature review and combining them alongside the analysis of the research statement and user interviews that were performed as well. All requirements were then compiled into a single document, the Software Requirements Specification document which would frequently be referred to during development.

REQ-1.5	
Description	When launching the app for the first time, guide the user by providing information on SM and SMA, helping the user set up a goal, and suggesting alternative activities to partake in.
Software Input	User launches the application for the first time, also includes textual user inputs when setting goals
Software Processing	Take the user inputted goal, and set it as the current goal until it is manually updated by the user, as well as any alternative activities the user can come up with
Software Output	Display a brief overview of the purpose of the app, alongside prompts for user input such as goals, and a user's own activity suggestions. Upon completion, display the main view of the app (see functional requirement 1.1)
Other	N/A
Constraints	User must not have completed the onboarding process / User must have never ran the app before
Data Handling	Similar to functional requirement 1.4, handle user input by restricting the character limit to a pre-determined amount
Error Handling	If the user closes the app during the onboarding phase, reset the onboarding phases to the first step; similar to if the user had opened the app for the first time again

Table 2: Functional Requirement 1.5

Table 2 shows an example of how a requirement was defined within the SRS. It contains a description of how it would function normally, the inputs required from an end-user to ensure its functionality, the processing required by the application, the output the app should show to an end-user after processing, constraints of the requirement, how the app should handle any data used for the requirement, and how the app should handle errors the requirement may be prone to.

Req. ID	Requirement Name	Requirement Description
REQ-1.1	View overall application usage	Allow the user to access a visual graph showcasing their app usage over the past week
REQ-1.2	Compare app usage from one week to the last	Allow the user to compare their app usage stats from one week to another, with the function also giving a percentage of the usage increase or decrease
REQ-1.3	Give guidance based on RT to the user if overall app usage is higher than last week's	If a user has seen a substantial increase in app usage from one week to another, provide step-by-step guidance to the user, similar to that seen when launching the app for the first time
REQ-1.4	Update current goal	Allow the user to update their current goal, originally defined when launching the app for the first time
REQ-1.5	Guide the user through setting up the app for continued use	When launching the app for the first time, guide the user by providing information on SM and SMA, helping the user set up a goal, and suggesting alternative activities to partake in
REQ-1.6	Send current usage statistics for future analysis	Allow the user to send their current usage statistics to the project leader for analysis and evaluation
REQ-1.7	Suggest alternative activities	Provide the user with alternative activities to do instead of using their smartphone / social media
REQ-1.8	Update current set of activities	Allow the user to update their current set of activities originally defined when launching the app for the first time.

Table 3: Functional Requirements Master List

Table 3 provides a full list of all the functional requirements identified throughout the project's lifespan. With the exception of "REQ-1.8", where the requirement was identified and defined later during development, all requirements were identified during the initial requirements analysis phase of development. A comprehensive breakdown of each functional requirement identified in detail, similar in manner to Table 2, alongside non-functional requirements can be found in Appendix F, as well as the SRS document separately via the GitHub repository.

3.4 Design

The next step was to design both design diagrams that represent how the requirement should typically function for an end-user and user interfaces for each screen necessary per requirement.

3.4.1 Diagram Design

The type of diagram used for designing the application based on the requirements was a Flowchart diagram, a diagram which uses components, typically shapes, that represent logic, order, and decision-points within a system to model said system or related processes, making it easy for developers to understand event sequences or the overall structure of a system .This type of diagram was decided upon due to how easily it can convey the idea of how a requirement should function . Flowchart diagrams were the exclusive diagram used for designing the application.

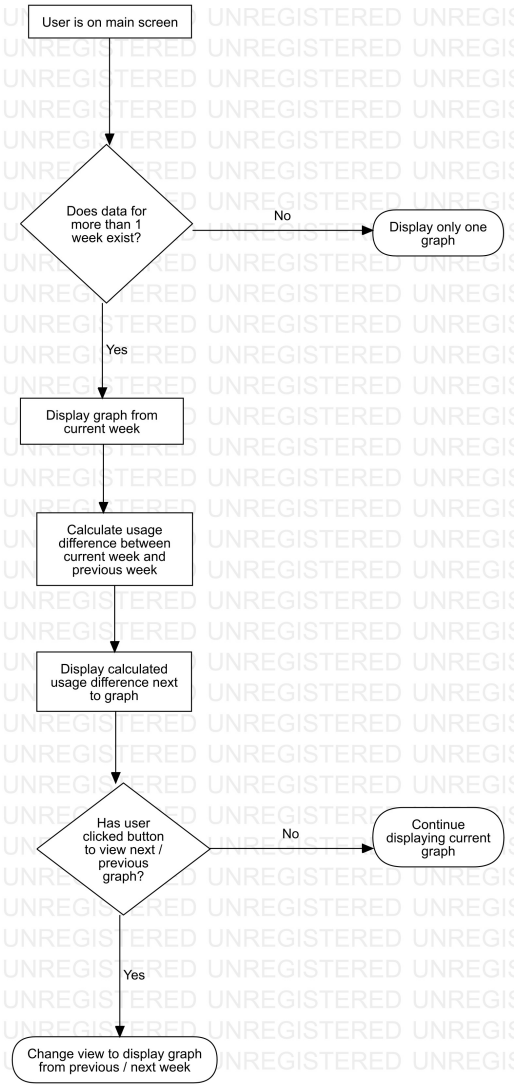

Figure 4: *Req-1.2 Flowchart Diagram*

Figure 4 provides an example of the flowchart diagrams created during the design stage. The oval shapes represents the start and end points for a diagram, the arrow lines represent the path taken by the program during execution, the diamond shapes represent decisions that are to be made by the app, and the rectangles represent processes during the execution of the app. Typically, more shapes are used to represent other parts of a system design, however, to keep the design as simple possible to understand, the prior three shapes were the only ones utilised during this process, with the level of complexity in the designs varying depending on if elements such as error handling were part of the requirement.

This process of creating a flowchart based on the requirement was repeated until all requirements outlined had a flowchart diagram. From there, the designs were reviewed for any possible refinements, for which only one flowchart needed refinement, that being requirement 1.3. Requirements were created in sequential order according to the requirement number (Req-1.1 → Req-1.2 → Req-1.3 → etc. . .). A complete list of the flowchart diagrams created can be found in Appendix G.

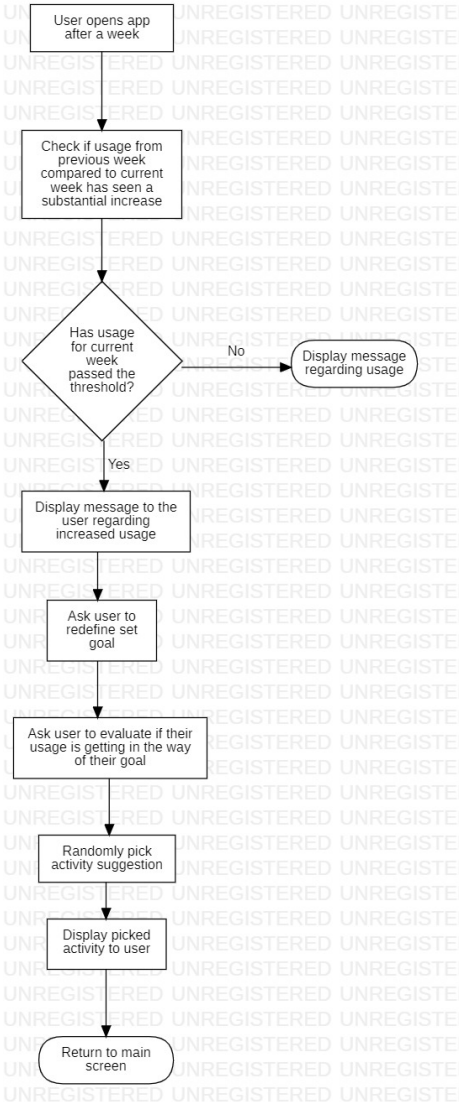

Figure 5: *First iteration of requirement 1.3's flowchart design*

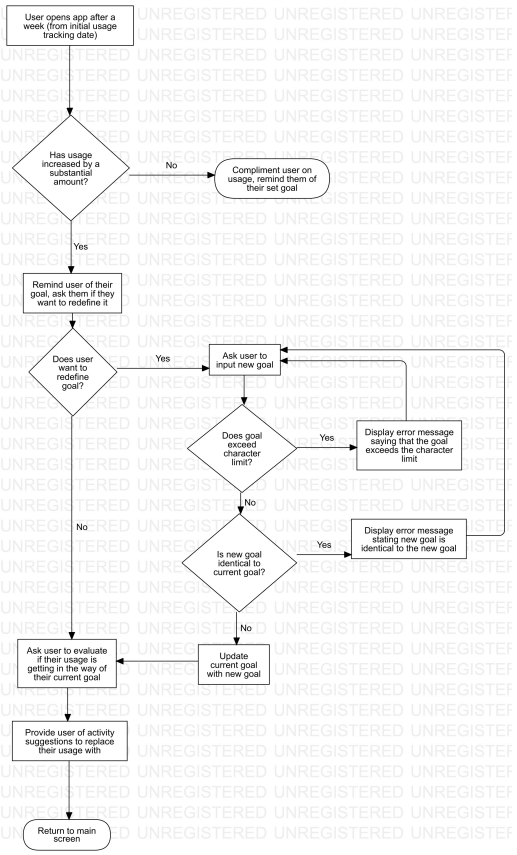

Figure 6: *Final iteration of requirement 1.3's flowchart design*

Figures 5 and 6 show the first and last iterations respectively of the flowchart design for requirement 1.3. While the design for the requirement is the same, the later iteration shows greater detail, providing more decision-points and error handling in the scenario the user wants to redefine their goal, while also making some changes to the wording of the process points.

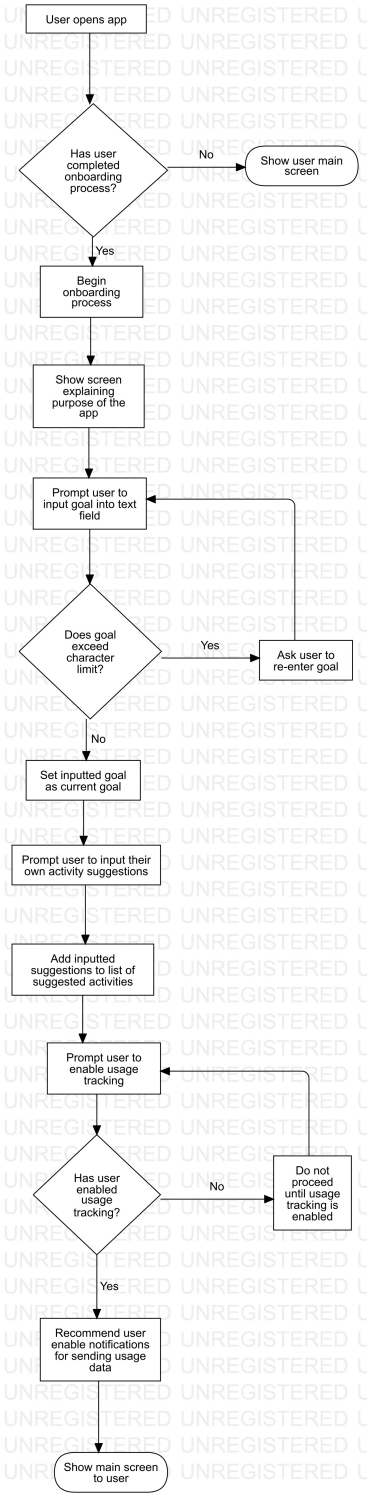

Figure 7: *Req-1.5 / Req-1.7 Flowchart Diagram*

Figure 7 shows the flowchart diagram of requirements 1.5 and 1.7, merged into a single requirement. When creating both requirements, it was identified that requirement 1.7 could only function if the app was at a specific point. Requirement 1.7's constraints notes that:

"User must be currently given guidance by the app, or is in the onboarding section of the app"

This means that the requirement can't be met unless the app is in the onboarding stage (outlined in requirement 1.5) or is currently providing guidance to the user (outlined in requirement 1.3). To avoid replication when creating the flowcharts, requirements 1.5 and 1.7 were merged into a single diagram.

3.4.2 User Interface Design

When designing the UI, a wireframing approach was used. This approach involves creating simplistic designs of an interface to establish structure and design flow. Wireframe designs are meant to be basic in terms of visual spectacle, acting as a blueprint for a user interface, and keeping its focus solely on fundamental design elements such as where important parts of a UI should go . The intention from the start was to keep the UI as simple as possible, ensuring that an end-user wasn't overwhelmed from using the app while making sure all essential elements for the interface were included, therefore, this approach was deemed to be the most effective way of creating the UI.

The wireframes created for the UI consist of "Mid-Fidelity" wireframes, which can be defined as a compromise between "Low-Fidelity" and "High-Fidelity" wireframes, providing clearer structure of an interface while making use of little to no visual flair or colouring.

Because all of the requirements had already been established (with the exception of requirement 1.8), there was a fairly clear understanding of what UI elements needed to be included to ensure functionality for each requirement. One screen by itself did not meet the functionality for each requirement, meaning the creation of multiple screens were necessary to ensure each requirement was met. Concerning how the screens needed for each requirement should be ordered, the flowcharts established how the screens should be ordered.

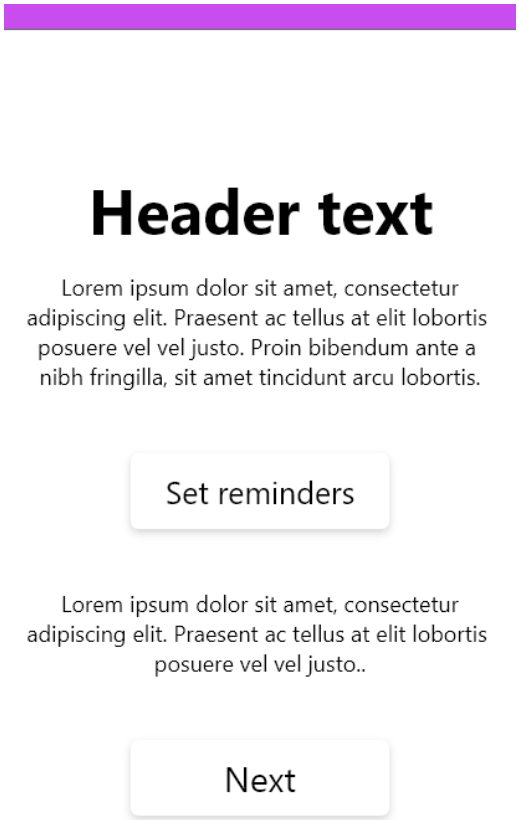

Figure 8: *UI wireframe of the "Enable Notifications" screen during the onboarding process - Made for Requirement 1.5/1.7*

An example of the UI interface design created can be seen in figure 8. The UI in this example seen showcases three core elements, the "Header Text" in large, bold text, denoting the heading for that screen, smaller text shown by the "Lorem ipsum" text, used for communication information to a user, and a button shown by the rectangle with the drop-shadow, denoting interactive elements for the screen. A full list of the screens created can be found in Appendix H.

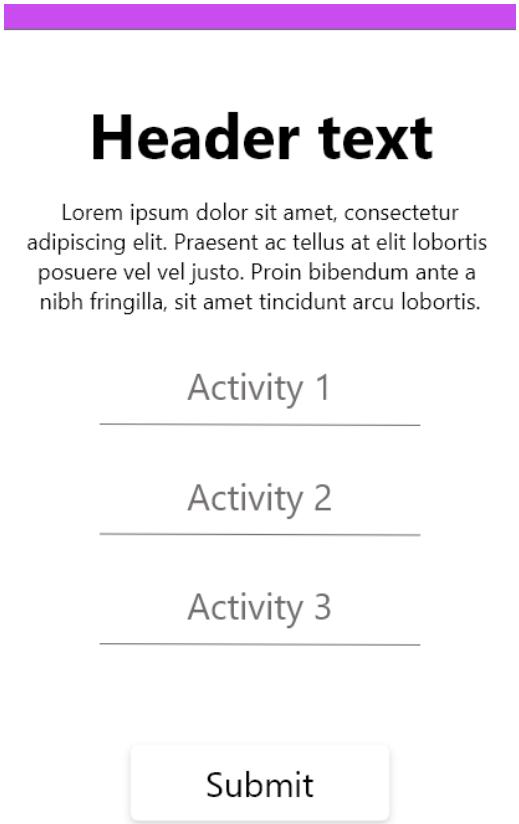

Figure 9: *UI wireframe of the "Activity Entry Screen" screen during the onboarding process - Made for Requirement 1.5/1.7*

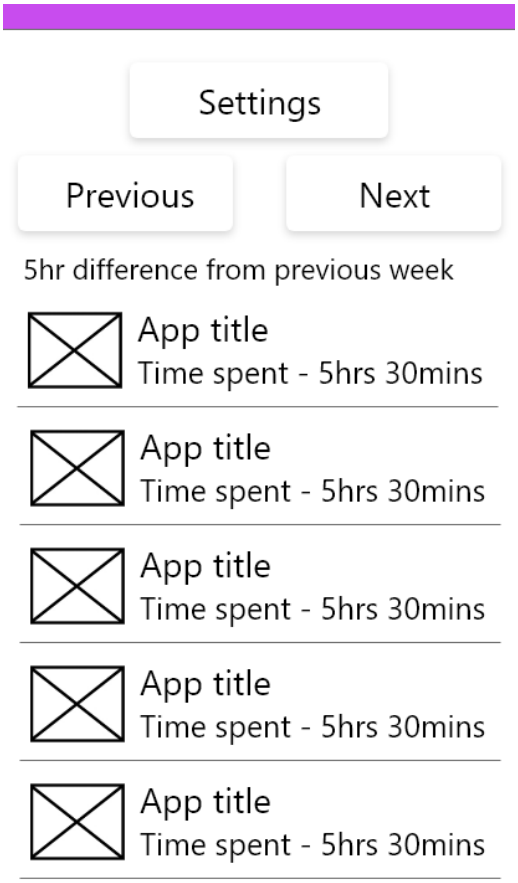

Figure 10: *UI wireframe of the Main Screen*

Each screen was varied in terms of interactive elements, text required, etc. Figures 9 and 10 show examples of different elements that a screen may include, such as lists or text entry fields.

While creating these screens, it was also determined that some of the screens can be reused in other requirements. For example, the screen used for entering a goal are needed in requirements 1.4 and 1.5, meaning time could be saved from not having to create another screen.

3.5 Implementation

At this stage, implementing the project would consist of using the information gathered for each requirement during the requirements analysis stage, then taking the flowchart diagrams and UI wireframes created from the design stage, and implementing them according to the documentation.

Figure 2 from section 3.3 provides the implication, however, it should be plainly stated that the requirements weren't implemented in sequential order, as the author found that it would prove troublesome attempting to implement some requirements before others

Rather, requirements were implemented in an order of what would be feasible to implement at that point in development.

3.5.1 Proof-of-Concept Prototype(s)

Before beginning work on any of the requirements, a subfolder that would contain a basic prototype for ensuring the UsageStatsManager class discussed in section 2.4 can return the data necessary for the app to function as intended and ensuring the app's notification system was created. This would serve as a blueprint from where the app would then be built upon.

```
public void loadUsage() throws PackageManager.NameNotFoundException {
    long lastWeek = System.currentTimeMillis() - (1000*3600*24*7);
    UsageStatsManager usm = (UsageStatsManager)this.getSystemService(USAGE_STATS_SERVICE);
    List<UsageStats> appList = usm.queryUsageStats(UsageStatsManager.INTERVAL_WEEKLY,
            lastWeek, System.currentTimeMillis());
    appList = appList.stream().filter(app -> app.getTotalTimeInForeground() > 0).collect(Collectors.
     toList());

    Log.d("usagestats", appList.toString());

    if (!appList.isEmpty()) {
        Map<String, UsageStats> sortedMap = new TreeMap();
        for (UsageStats usageStats : appList) {
            sortedMap.put(usageStats.getPackageName(), usageStats);
        }
        Log.d("moreStats", sortedMap.toString());
        showUsage(sortedMap);
    }

}
```

Code 1: loadUsage() function

Code snippet 1 shows the function used to fetch the usage data for the user. This function is called on app startup.

The function calls the "queryUsageStats" method from the UsageStatsManager class to get the usage stats from the last week, then stores it in a list, then, if the list isn't empty, maps it to a tree map then calls another function "showUsage" to show the usage to the user. Some of the code seen above was sourced from an external

sample project, see Appendix I for more information.

After ensuring that all the necessary information was able to be fetched from the UsageStatsManager class, a separate folder was created that focused on fully developing the app. The folder structure can be seen below in figure 11.

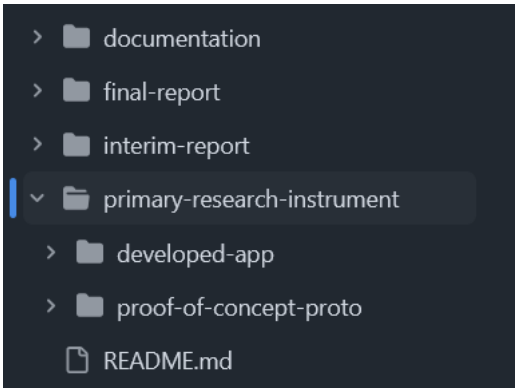

Figure 11: *Folder structure of GitHub repo*

This is where a secondary prototype was created, used for ensuring the app was able to send notifications reminding the user to send their usage statistics was able to be implemented. As the evaluation involved testing the application with human participants, implementing a reminder system using notifications was thought to be a necessary feature to ensure the data is sent in a timely manner.

```
final PeriodicWorkRequest periodicWorkRequest =
        new PeriodicWorkRequest.Builder(WorkerClass.class, 5, TimeUnit.SECONDS, 15, TimeUnit.MINUTES
    )
        .addTag("periodicWork")
        .build();

notiTestBtn.setOnClickListener(view ->
        WorkManager.getInstance().enqueue(periodicWorkRequest));
```

Code 2: PeriodicWorkRequest class used for sending notifications

```
public class WorkerClass extends Worker {

    public WorkerClass(@NonNull Context context, @NonNull WorkerParameters workerParams) {
        super(context, workerParams);
    }

    @NonNull
    @Override
    public Result doWork() {
        try {
            Thread.sleep(1000);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
```

```
        sendNoti();

        return Result.success();
    }
    public void sendNoti() {
        NotificationManager notiManager = (NotificationManager)getApplicationContext().
     getSystemService(Context.NOTIFICATION_SERVICE);
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            CharSequence name = "FocusathChannel";
            String desc = "App notification channel for Focusath";
            int importance = NotificationManager.IMPORTANCE_DEFAULT;
            NotificationChannel channel = new NotificationChannel("default", name, importance);
            channel.setDescription(desc);
            //NotificationManager notificationManager = getSystemService(NotificationManager.class);
            notiManager.createNotificationChannel(channel);
        }

        NotificationCompat.Builder notification = new NotificationCompat.Builder(
     getApplicationContext(), "default")
                .setContentTitle("title")
                .setContentText("message")
                .setSmallIcon(R.mipmap.ic_launcher);
        notiManager.notify(1, notification.build());
    }

}
```

Code 3: WorkerClass class used for sending notifications

Code snippet 2 and 3 shows the code and class used for sending notifications. It defines a PeriodicWorkRequest variable that triggers the code in code 3 every 5 seconds, 15 minutes or later from the last trigger, then, creates an OnClickListener that queues the PeriodicWorkRequest. Code snippet 3 executes the doWork() method which calls sendNoti() function. This builds the notification to be sent to the user, before displaying it to the user. Seeing as the code in both snippets were for ensuring functionality, both would be refined over the course of development.

This prototype would also establish the class structure for the app, as can be seen in figure 12 below. By the end of development, RecyclerViewAdapter would be the only class removed.

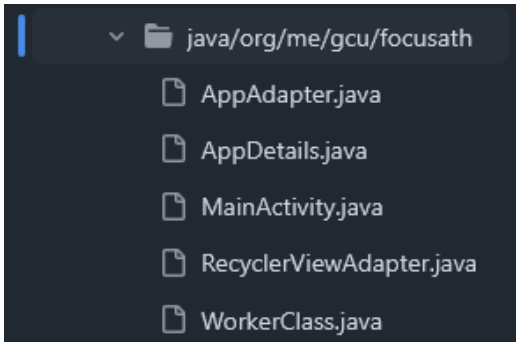

Figure 12: *Class structure of PRI at the prototyping stage*

3.5.2 Req-1.5 / 1.7

This branch focused on creating the onboarding process, the process the user goes through when opening the app for the first time, and expands on the groundwork established with the previous prototype. Development with regards to the SRS began with these requirements as this would establish the core functionality of the app.

```
isOnboardingComplete = sharedPreferencesOnBoarding.getBoolean("isOnboardingComplete", false);
    if (!isOnboardingComplete) {
        if (screen_count > 1) {
            sharedPreferencesOnBoarding.edit().putBoolean("isOnboardingComplete", false).apply();
            screenCheck();
        } else {
            sharedPreferencesOnBoarding.edit().putBoolean("isOnboardingComplete", false).apply();
            screen_count = 1;
            screenCheck();
        }
    }
    else {
        if (getGrantStatus()) {
            screen_count = 0;
            showStatsBtn.setOnClickListener(view -> {

            });
        }
    }
```

Code 4: Checks if user has completed onboarding process

Code snippet 4 shows the code used for checking if the user has completed the onboarding process, and is called every time the user opens the app. The code uses a "SharedPreference", an API used to store key-value data, even after the app is closed. At the end of the onboarding process, the SharedPreference for the onboarding process is set to true, so when the app is opened, the skips the initial check, and then checks if the app has been granted access to usage statistics.

```
public void screenCheck() {
    //TODO: move onboarding usage tracking code to here
```

```
    if (screen_count == 0) {
        setContentView(R.layout.activity_main);
        try {
            loadUsage();
        } catch (PackageManager.NameNotFoundException e) {
            throw new RuntimeException(e);
        }
    }
    //note: this has been moved to be the first screen as exiting the intent returns to the first
     screen, TODO: update uml flowchart to reflect change
    else if (screen_count == 1){
        setContentView(R.layout.enable_user_tracking_screen);
    //...
    }
    //...
}
```

Code 5: Code used for swapping between screens

Code snippet 5 shows an example of the code used for swapping between screens. A screen is linked to a corresponding number, and whenever, for example, a button is clicked, the code calls the method screenCheck(), then swaps to the corresponding screen. The code snippet shown has been shortened, due to the function being particularly large, but the full function can be found in Appendix J.

Initially, this method of swapping between screens was used when debugging the application, but this method of swapping between screens was used for the remainder of development for simplicity purposes.

A constraint for requirement 1.5 defined in the SRS (see Appendix F) stated that characters entered by the user should be restricted to a pre-determined amount. This is achieved in the XML file seen in code snippet 6 below where the structure for the goal entry screen is defined. XML files in android app development define the layout for a UI, and house components such as buttons and text, referred to as widgets. The parameter "android:maxLength" sets the max length that can be entered in an input box.

```
 <EditText
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/goalEntryField"
    android:autofillHints="goalEntry"
    android:maxLength="25"
    android:gravity="center_horizontal"
    android:inputType="textNoSuggestions|textVisiblePassword"
    android:hint="Type in a goal"
    android:ems="10">
</EditText>
```

Code 6: XML code for the EditText widget

3.5.3 Req-1.1

This branch focused on building the graph showing the user their app usage. Figuring out how to display the graph wasn't an issue, as this was already achieved in the prototype discussed in section 3.6.1, so this branch was used to adapt the screen to match the UI wireframe shown in figure 10.

When displaying the graph in the UI, the app uses a widget called a ListView whose contents can be adapted to show what a developer wants. In this case, the contents of each item in the lists consist of the icon of an app, the app name, the usage time measured in hours and minutes, and the usage percentage. All of the details are populated using the code seen in code snippet 1.

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical">
    <LinearLayout
        android:layout_height="50dp"
        android:scaleType="centerCrop"
        android:id="@+id/appIcon"/>
        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="5sp"
            android:layout_marginTop="15sp">
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="AppName"
                android:textSize="15sp"
                android:textStyle="bold"
                android:id="@+id/app_name_text"
                android:gravity="center_horizontal"/>
            <View
                android:layout_width="30dp"
                android:layout_height="1dp"
                android:layout_marginTop="10sp"
                android:layout_marginStart="10sp"
                android:background="@android:color/darker_gray"/>
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="0h"
                android:textSize="15sp"
                android:textStyle="bold"
                android:layout_marginStart="10sp"
                android:id="@+id/usage_time_text"
                android:gravity="center_horizontal"/>
            <View
                android:layout_width="30dp"
                android:layout_height="1dp"
                android:layout_marginTop="10sp"
                android:layout_marginStart="10sp"
                android:background="@android:color/darker_gray"/>
            <TextView
                android:layout_width="wrap_content"
```

```
                android:layout_height="wrap_content"
                android:text="50\%"
                android:textAlignment="textEnd"
                android:textSize="15sp"
                android:textStyle="bold"
                android:layout_marginStart="10sp"
                android:id="@+id/usage_percent_text" />
    </LinearLayout>
</LinearLayout>
```

Code 7: XML code used by the ListView widget for displaying code

Code snippet 7 shows the XML code the ListView widget uses when displaying the usage statistics code. It consists of TextView widgets that are used as placeholders that are then filled in by the actual values.

3.5.4 Req-1.4

This branch focused on adding functionality for the user to change their goal via the settings screen. Because the goal entry screen was already created previously for requirement 1.5/1.7, the XML code was simply recycled and refined slightly, therefore most of the coding consisted of creating the UI for the settings screen, and adding input validation for the goal re-entry screen.

```
 submitNewGoalBtn.setOnClickListener(view -> {
    String oldGoal =  sharedPreferences.getString("goalEntry", "none");
    newGoalEntryText = newGoalEntryField.getText().toString();
    if (newGoalEntryText.isEmpty()) {
        Toast.makeText(view.getContext(), "Please enter a goal to continue", Toast.LENGTH_SHORT).
     show();
    } else if (oldGoal.equals(newGoalEntryText)) {
        Toast.makeText(view.getContext(), "New goal cannot be identical to old goal", Toast.
     LENGTH_SHORT).show();
    }
    else {
        sharedPreferences.edit().putString("goalEntry", newGoalEntryText).apply();
        Toast.makeText(view.getContext(), "Goal has been updated to: " + newGoalEntryText, Toast.
     LENGTH_LONG).show();
        screen_count = 7;
        screenCheck();
    }
});
```

Code 8: Code for goal re-entry screen

Code snippet 8 shows the code for the goal re-entry screen. When the button to submit a new goal is pressed, the code checks if the text entry field is empty or is the same as the current goal, and displays a message accordingly. If the text entered passes both checks, it updates the goal via the previously discussed SharedPreference API, then displays a message notifying the user about the update, then goes back to the settings screen using the screenCheck() function.

3.5.5 Req-1.6

This branch focused on allowing a user to draft an email containing their usage stats for the week, as outlined in requirement 1.6. There was one significant issue encountered during the implementation of requirement 1.6. While opening the email client of a user wasn't an issue, the actual issue came from attempting to attach the .csv file to the email draft.

```
private void writeAppListToFile (ArrayList<AppDetails> appStuff) {
    try {
        FileOutputStream fileOutputStream = openFileOutput("myfile.txt", Context.MODE_PRIVATE);
        ObjectOutputStream outputStream = new ObjectOutputStream(fileOutputStream);
        for (int i = 0; i < appStuff.size(); i++) {
            outputStream.writeObject("\n" + appStuff.get(i).appName + "  |  " + appStuff.get(i).
     usageTime);
        }
        //outputStream.flush();
        outputStream.close();
        fileOutputStream.close();
        Log.d("fileDebug", "app details written to file!");
    } catch (IOException e) {
        throw new RuntimeException(e);
    }
}
```

Code 9: Original implemention of writing the app list to a file

Shown above in code snippet 9 is the code used to write the usage data to a file. Originally, the usage data file is stored in the app's internal storage, and while this successfully wrote the usage data to the text file, this created an unforeseen consequence, that being the email client couldn't access the file, and therefore couldn't attach the file.

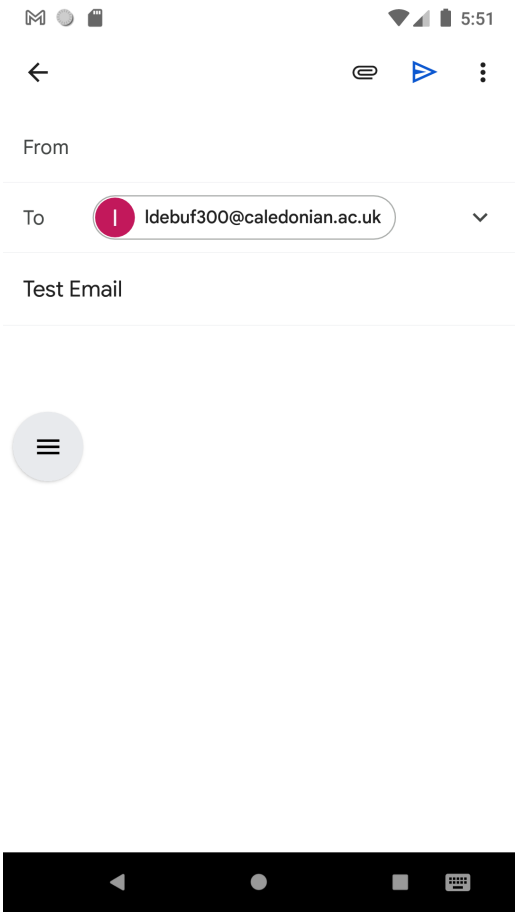

Figure 13: *File attachment issue (email recipient has been hidden)*

This required rewriting a majority of the function used to create the file. Instead of storing the file in internal storage, the file now had to be stored in a directory inside external storage, and instead of using the FileOutputSteam class, which is better suited for multimedia files, the function now uses the FileWriter class, better suited for textual data . This rewritten code, and the successful attachment is shown below in code snippet 10 and figure 14.

```
private void generateFile(ArrayList<AppDetails> appDetails) {
    String fileName = "appDataDetails.csv";
    File exFileDir = this.getExternalFilesDir(Environment.DIRECTORY_DOCUMENTS);
    fileToEmail = new File (exFileDir, fileName);
    try {
        FileWriter fileWriter = new FileWriter(fileToEmail);
        for (int i = 0; i < appDetails.size(); i++) {
            fileWriter.append("\n").append(appDetails.get(i).appName).append("  |  ").append(
     appDetails.get(i).usageTime);
        }
        fileWriter.flush();
        fileWriter.close();
```

```
        Log.d("fileStuff", "File created successfully, located at: " + exFileDir);
    } catch (IOException e) {
        e.printStackTrace();
    }
}
```

Code 10: New implemention of writing the app list to a .csv file

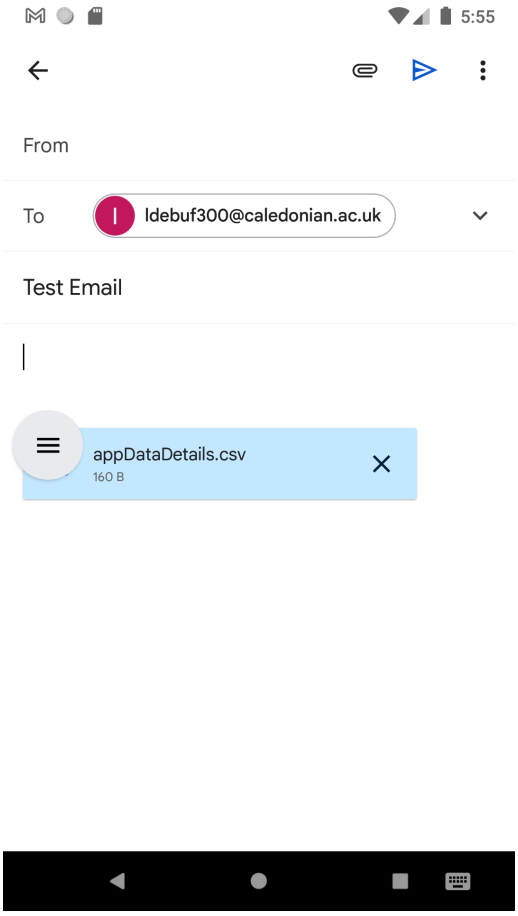

Figure 14: *File attachment issue, now fixed (email recipient has been hidden)*

3.5.6 Req-1.3

This branch is where the WDEP system from RT mentioned in section 2.3.3 of the literature review would see its continued implementation. The first step in implementing this was to create conditions the code can reach to show separate screens depending on if a user had seen an increase or decrease in their usage over the last week. This involved figuring out a way to store a users old usage time to compare, as the comparison would be used to send the user to correct set of screens.

```
long currentTime =  TimeUnit.MILLISECONDS.toSeconds(totalTime);
long oldTime = TimeUnit.MILLISECONDS.toSeconds(sharedPreferencesUsageTime.getLong("oldUsageTime", 0)
     );

if (currentTime > oldTime) {
    Log.d("changeScreen1", "screen time has increased, different screen will appear here");
    screen_count = 10;
    screenCheck();
} else {
    Log.d("changeScreen2", "screen time has droppped, different screen will appear here");
    screen_count = 9;
    screenCheck();
}

sharedPreferencesNotiSent.edit().putBoolean("notiSent", false).apply();
sharedPreferencesUsageTime.edit().putLong("oldUsageTime", totalTime).apply();
```

Code 11: Checking for usage increase / decrease

Code snippet 11 shows the code used when the app checks for a usage increase or decrease, called once a week. currentTime is defined as the time value retrieved from the loadUsage() function (see code snippet 1) while oldTime is the stored SharedPreference value of the variable, last updated from the prior week, when the function was last ran. The code will then swap to a corresponding screen depending on if there was an increase or decrease in usage. The code uses seconds as its time unit for debugging, while the final product makes use of hours instead.

Because the user has the option of redefining their goal if they have seen a usage increase, the code for defining a goal had to be rewritten slightly. Shown below in code snippet 12, the code now checks if boolean value isRedefiningGoal is true, and if it is, updates the entered goal from the user.

```
 else if (isRedefiningGoal == true) {
    //TODO: possibly change paragraph text to make more sense given the context
    String oldGoal =  sharedPreferences.getString("goalEntry", "none");
    if (oldGoal.equals(goalEntryText)) {
        Toast.makeText(view.getContext(), "New goal cannot be identical to old goal", Toast.
     LENGTH_SHORT).show();
    }
    else {
        sharedPreferences.edit().putString("goalEntry", goalEntryText).apply();
        Toast.makeText(view.getContext(), "Goal has been updated to: " + goalEntryText, Toast.
     LENGTH_LONG).show();
        screen_count = 11;
        screenCheck();
        isRedefiningGoal = false;
    }
}
```

Code 12: Goal definintion screen, rewritten for requirement 1.3

Code 13 below shows the code used for displaying a set of activities from the user. While the user activities are simply retrieved from the stored value using the SharedPreference API, the random hard-coded activities are retrieved from a sepa-

rate string array file called strings.xml (see Appendix K), then two pseudorandom numbers are selected and are used to pick a random activity.,

```
String[] activities = getResources().getStringArray(R.array.activities_array);
Random random = new Random();
int randomNumberOne = random.nextInt(activities.length - 1);
int randomNumberTwo = random.nextInt(activities.length - 1);

String randomActivityOne = activities[randomNumberOne];
String randomActivityTwo = activities[randomNumberTwo];

String userActivityOne = sharedPreferences.getString("activityOne", "none");
String userActivityTwo = sharedPreferences.getString("activityTwo", "none");
String userActivityThree = sharedPreferences.getString("activityThree", "none");

activitiesText.setText(userActivityOne + ", " + randomActivityOne + ", " + userActivityTwo + ", " +
     randomActivityTwo + ", " + userActivityThree);
```

Code 13: Code used for displaying activities to a user

3.5.7 Req-1.2

This branch was used to add

the graph used for displaying the usage for the previous week to the user. While a wireframe diagram for this screen was never created, re-using the idea of displaying the data in a list format was the decision made.

The next step was establishing a method of storing the prior week's usage data to show the user. Because the code for creating a file already existed (see snippet 10), the code was simply adapted to create a separate .csv file if there was already an existing file. This adapted code can be seen below in code snippet 14

```
if (notiSent) {
    if (fileToEmail.exists()) {
        Log.d("fileExists", "file " + fileName + " already exists, creating file for previous week")
     ;
        String fileNamePrevWeek = "appDataDetailsPrev.csv";
        File exFileDirPrevWeek = this.getExternalFilesDir(Environment.DIRECTORY_DOCUMENTS);
        fileToEmail = new File (exFileDirPrevWeek, fileNamePrevWeek);
        //...
    }
}
```

Code 14: Adapted code for storing usage data

The data from the .csv file then needs to be parsed properly to display it to the user, and this was achieved by using the external library OpenCSV abd the CSVReader class. The code shown below in snippet 15 reads the file line-by-line, then adds it to an array list. Similar to what was achieved for requirement 1.1, this array list is then added to a ListView widget, then shown to the user. The full code block can be found in Appendix L.

```
CSVReader csvReader = new CSVReaderBuilder(new FileReader(fileToShow)).withSkipLines(1).build();
String[] nextLine;
while ((nextLine = csvReader.readNext()) != null) {
    Log.d("prevWeek", nextLine[0]);
    previousWeekArrayList.add(nextLine[0]);
}
```

Code 15: Code used for reading .csv file data

3.5.8 Req 1.8 / Clean up

This was the last branch created during development, and focused on getting the app ready for human evaluation. This branch was also when the last requirement, requirement 1.8, was identified and implemented. Implementing requirement 1.8 was conducted in a similar manner to requirement 1.3, that being to add a boolean variable, check if its true, perform input validation, then update the set of activities. A small snippet of the code used can be found below in code snippet 16, but the full code block where this was performed can be found in Appendix M.

```
 if (isEditingActivities == true) {
    String activityOne = sharedPreferences.getString("activityOne", "none");
    String activityTwo = sharedPreferences.getString("activityTwo", "none");
    String activityThree = sharedPreferences.getString("activityThree", "none");

    activityFieldOne.setText(activityOne);
    activityFieldTwo.setText(activityTwo);
    activityFieldThree.setText(activityThree);

}
//...
 /// if currently editing activities, send toast indicating change, then return to settings screen
if (isEditingActivities == true) {
    Toast.makeText(view.getContext(), "Activities have been updated", Toast.LENGTH_LONG).show();
    screen_count = 7;
    screenCheck();
    isEditingActivities = false;
    return;
}
```

Code 16: Code added to activities definition code block

Aside from adding this functionality, this branch replaced all of the placeholder text from previous branches with actual text, most of which was added to the strings.xml file mentioned in section 3.6.6, added some extra functionality not explicitly noted in the SRS such as a help alert on the main screen, adjusted some of the time-sensitive values to reflect the intended values, and added an app logo.

3.6 Testing

At the end of each prototype iteration, Incremental Integration Testing was carried out to ensure that the requirements were implemented as intended according to how the requirement was defined in the SRS. Incremental Integration Testing can be defined as testing components of a software system, one by one, adding components together until eventually the entire system is being tested. As the nature of development involved creating a component based on the requirements specified, and only continuing once the component had been created to match the specification, this method of testing felt most appropriate.

Test ID	**Action**	**Expected Outcome**	**Actual Outcome**	**Evidence**	Assumptions / Context (If necessary)
2.1	User clicks on "previous week" button while no data for the previous week exists	A dialog pops up explaining to the user that no data for the previous week currently exists, and that they should check back later	A dialog pops up explaining to the user that no data for the previous week currently exists, and that they should check back later	Req_2.1.webm	

Table 4: Example of a test case

An example of how testing was carried out can be seen in Table 4. The test case includes the test ID, mapped to match the requirement number, the action taken by a user that would trigger the component, the expected outcome of the test, the actual outcome of said test, evidence to show the outcome of the test, usually in the form of a screen recorded video or screen capture, and any additional context or assumptions needed to understand the test case. A comprehensive list of the test cases and their results can be found in Appendix N.

4 Evaluation & Results Discussion

This section provides details on the evaluation of the project in relation to the hypothesis stated in section 1.3, which was:

The combination of usage tracking and intervention methods rooted in psychology within the created mobile application will help to address and reduce smartphone & social media usage in young adults.

It also gives critical analysis and discussion of the results and findings from the evaluation, before drawing initial conclusions based on the project hypothesis, research statement and objectives. Given the aforementioned hypothesis, the intention of the evaluation was to find if:

1. The mobile phone usage from a participant decreased when comparing their usage from the start of the evaluation to the end.
2. If there was a decrease in usage, was the PRI actually a contributing factor to it?

When evaluating if the PRI met the project's hypothesis, objective, and research statement, a mixed-method approach was used. This approach can be defined as utilising quantitative and qualitative methods in a study to obtain a deeper understanding of a phenomenon . The evaluation results are based off of the captured usage data from each participant, as well as the participant's own opinions of their experience using the PRI via a survey given at the end of the evaluation period.

4.1 PRI-Evaluation results

Participants who agreed to take part in the evaluation were given the PRI for a period of 3 weeks.

Once they had set up the application, they were also asked to send their usage immediately afterwards. This was to use said usage stats as a baseline to compare the usage for the next three weeks. While the data extracted initially shows the total usage which includes all apps used by a participant, the data was then further parsed by the author to analyse if there was a shift in usage of social media apps. As discovered in section 2.2 in the literature review, noted that people use their smartphones for productivity reasons, and not just for social media.

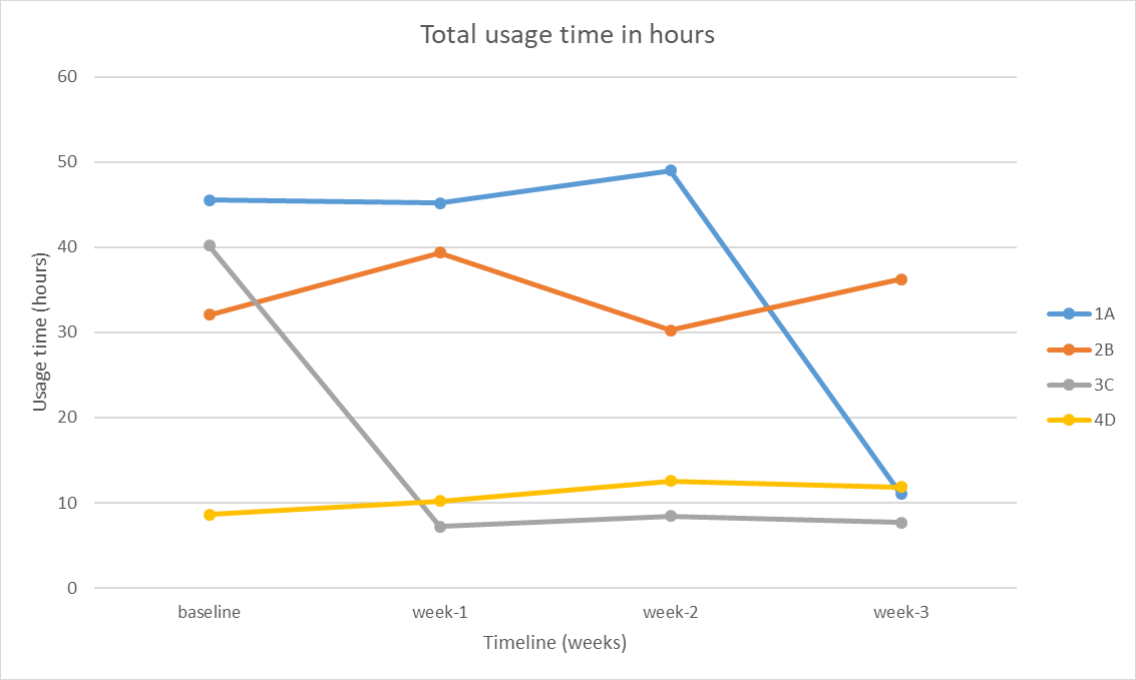

Figure 15: *Results from PRI evaluation*

Figure 15 shows the initial results from the evaluation, with the total usage from each participant. The x-axis shows the timeline of the evaluation period, from baseline to week 3. The y-axis shows the usage in hours. Table 5 shows a sample of the usage data captured, showing the name of the app package (a unique identifier for an android app), the usage time for the app, and the percentage of the usage time.

android	24:50:00	61%
maps	04:18	10%
One UI Home	01:21	3%

Table 5: Sample of usage data from participant 3C.

Within the dataset, an outlier was detected within the baseline data for participant 3C. Seen above in table 5, the "android" package, which refers to the android home screen, takes up over 24 hours of usage time. Seeing as this may be unrealistic for a typical user, the decision was made to leave out the usage time for "android". Figure 16 below shows the same chart as figure 15, but without the outlier data.

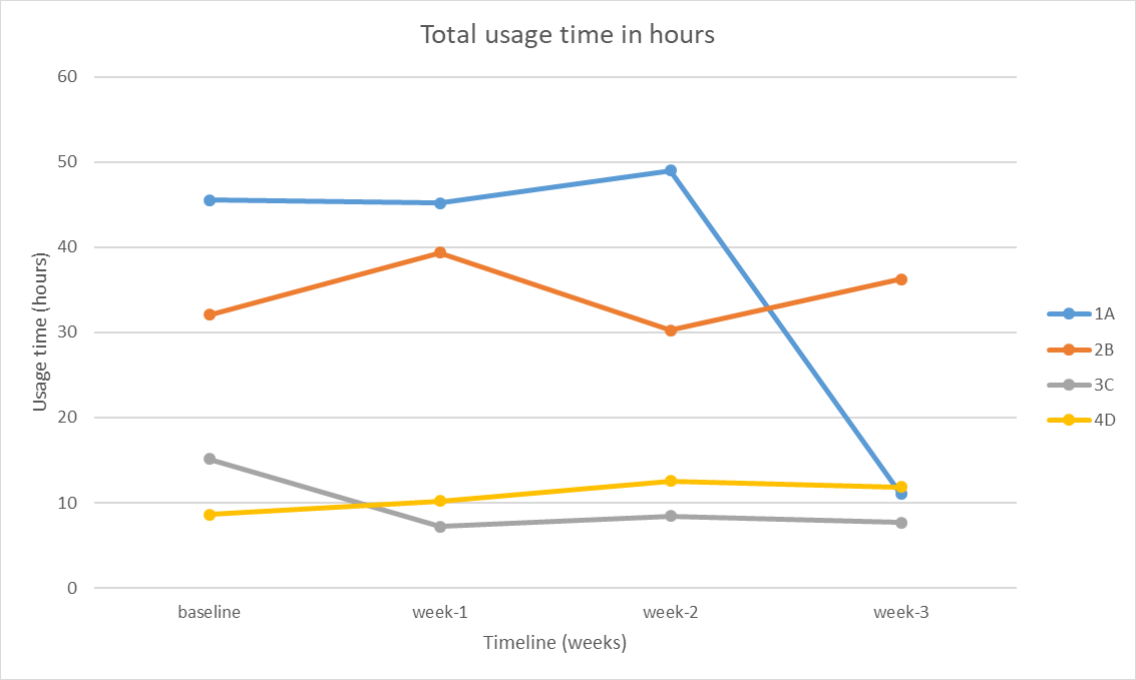

Figure 16: *Results from PRI evaluation, without the outlier data*

1A's usage is relatively consistent between baseline and week-2, though doesn't take a noticeable drop, but takes a substantial drop in week-3. 2B's usage fluctuates a lot throughout the timeframe, initially rising before falling again, and then settling in the middle for week-3. 3C's usage does take a drop overall when comparing baseline to week-3, and remains under 10 hours for the remainder of the evaluation period after the baseline data. 4D's usage experiences a slight increase before dropping slightly, though experiences an increase when compared from the baseline to week-3.

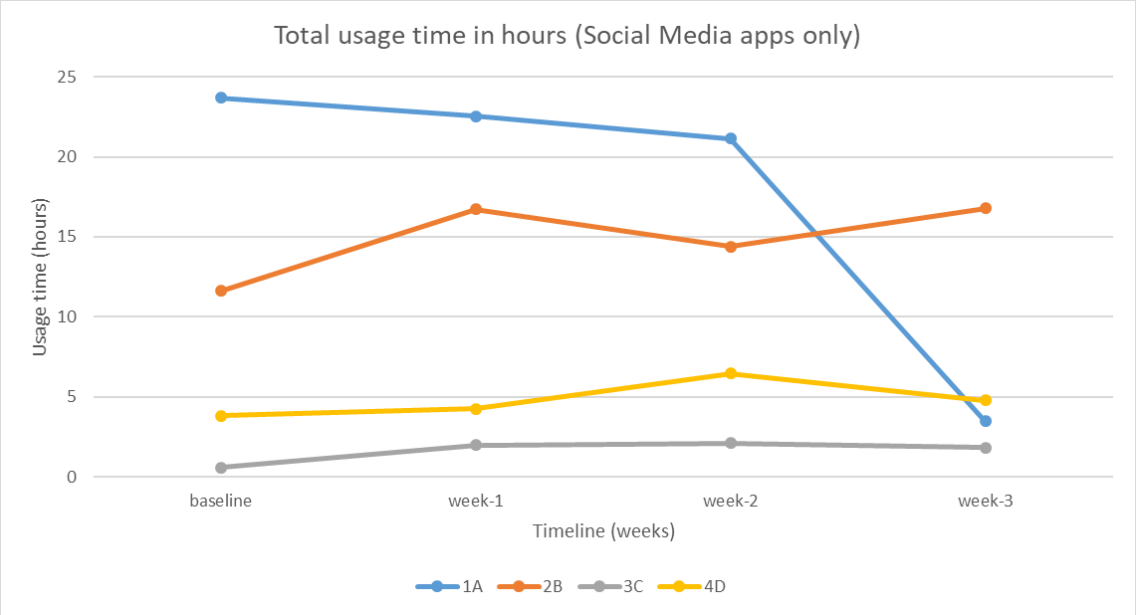

Figure 17: *Results from PRI evaluation, only showing social media usage*

Figure 17 shows the same graph as figure 16, but only including social media usage from each participant. While the graph is relatively the same, there are some differences. For example, unlike the total usage, 1A's social media usage never increases, rather, taking a small decline instead. 4D's social media usage starts, and remains quite low, though does take a small increase in week-1 before levelling out by the latter weeks.

4.2 Post-Evaluation survey results

Part of using a mixed-method approach involves collected qualitative data, which can be defined as written text, images, audio recordings to name a few . While the data from the previous section are an important piece of the evaluation as a whole, getting this kind of data helps to understand the thoughts and feelings from participants, and help in understanding and contextualising the behaviours represented in the previous section.

At the end of the evaluation testing period, participants were sent a survey to fill out. This survey gave participants the opportunity to express their opinions on using the app throughout the evaluation period, with the survey including a variety of multiple choice and open-ended, optional questions. Two question sets were provided, one focusing on how the participant felt after using the app, and the other focusing on the overall functionality of the app. The number of respondents

matched the number of participants who took part in the evaluation, that being 4. Note that because the open-ended questions were optional, not all participants provided a response to them.

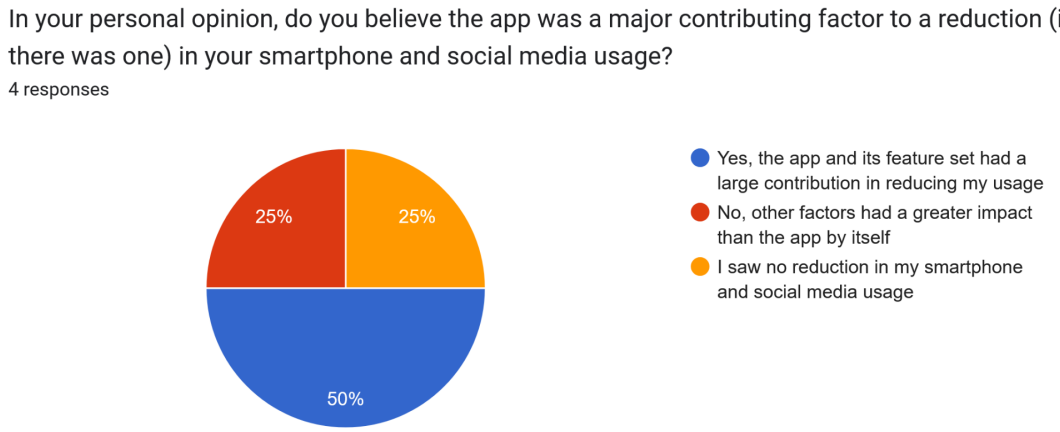

Figure 18: *Results from question 3 from the post-evaluation survey*

A complete collection of the responses and their associated questions can be found in Appendix O, however, the sentiment derived from the results of QS1 show that users felt more informed about their usage after using the app, and on average, users felt as if the app implemented the WDEP system to a good degree. As demonstrated in figure 18, only two of the respondents felt that the app contributed to their lowered usage, with one respondent feeling as if other factors had a bigger impact than the app by itself, and the other feeling like there wasn't any reduction in their usage. When asked to provide a reason to their choice, one respondent said in reference to the app lowering their usage:

"It works pretty well! I feel as if seeing your values and hobbies increases motivation to not chase social media and other applications"

Another response, in reference to the participant not seeing a usage drop, stated:

"At this time of year as a student based outside the city of my university I was regularly on my device to check on my classes and assignments"

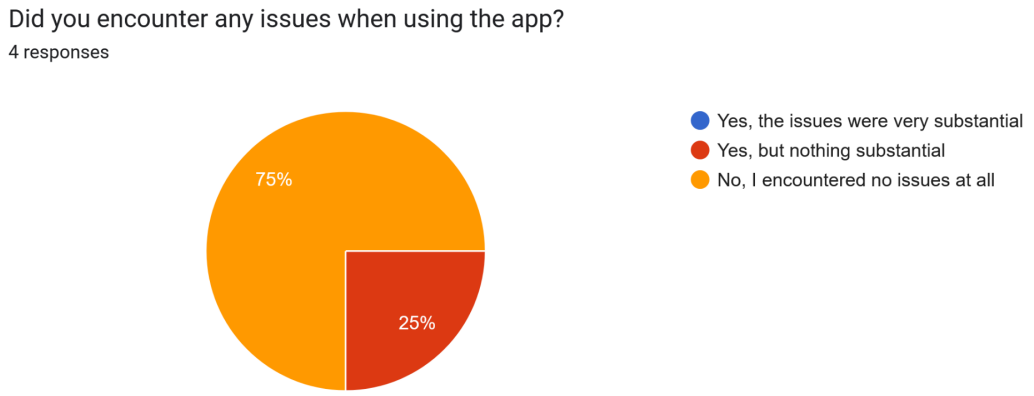

Figure 19: *Results from question 11 from the post-evaluation survey*

The responses to QS2 show that respondents generally had a positive experience using the app, with no participant experiencing any substantial issues. One participant did however note one smaller issue, stating:

"I find that the email prompt would be better and easier to understand is [sic] it was titled "report" or something similar"

Participants were asked if the app experience was better than existing apps that provided similar functionality, while 3 of the participants stated that they haven't used apps similar to the PRI, one participant disagreed that the app experience was better than existing apps. When asked to provide a follow-up stating what the PRI could take note of, they stated:

"an app called Screen Zen which gives more options to limit screen usage."

When asked to provide suggestions on how the app could be improved, one response suggested increasing the notification frequency from being sent on a weekly basis to a daily basis, while another suggested showing how the app usage for specific apps have changed between weeks. At the end, participants were finally asked to provide any closing thoughts regarding their experience, and while only one participant gave a response, they stated that they believe the app is a step in the right direction in addressing SA and SMA.

4.3 Discussion

The intention of the evaluation was to find if the PRI satisfies the project's hypothesis, and this was done by collecting quantitative data in the form of usage data captured from each participant and qualitative data in the form of a survey given to participants at the end of the evaluation period, where participants could provide their opinions on the current state of the PRI.

The quantitative data initially revealed that only three participants experienced a drop in smartphone usage at some point during the evaluation, and only two of the participants' usage was lower by the end of the evaluation period. Discussed in section 2.2, discussed how smartphones can be used for productivity and not exclusively for social media, so the data was reanalysed to look exclusively at social media apps, and found the results were largely the same between all participants when compared to the initial set of data.

The qualitative data revealed that participants found themselves informed about their smartphone and/or social media usage. This was another talking point from , as it was discussed that current apps don't provide an opportunity for users to reflect and learn about SA and SMA. Not only this, the data revealed that some participants attributed the PRI as a factor in their reduction in smartphone and social media usage, all participants in their opinion felt that the app helped them achieve the goal they set out, and most participants said they would recommend the app in its current state to others (see Appendix O), with one participant stating the app is a step in the right direction when addressing the issue of SA and SMA.

Given this, the PRI can be considered somewhat successful in meeting the project's hypothesis. This is because The quantitative data provided mixed results when it came to reducing the usage of all participants in both a general sense and specifically with social media, however, the qualitative data provided a juxtaposition, with some participants believing the app was helpful in reducing their usage, most saying they would recommend the app to a friend, and all participants believing the app helped them achieve their goals (see Appendix O).

5 Legal, Social, Ethical, and Professional Issues

This section will provide commentary on Legal, Social, Ethical, and Professional (LSEP) issues the project dealt with over its lifespan. Within the Social Issues section, it will also provide additional commentary and reflection on the Sustainable Development Goals (SDGs) the project has aligned itself with in accordance to the definitions provided by the United Nations.

5.1 Legal Issues

While the project didn't encounter any issues related to copyright, the project did have to deal with issues to data protection, particularly in relation to the Data Protection Act 2018 and the UK General Data Protection Regulation, two laws that relate to protecting the personal data of individuals .

When a human participant was involved at any stage of the project, it was clearly outlined to them what kind of data would be collected and processed. For example, during the evaluation stage, when the user opens the app for the first time, the user is given full clarity on the purpose of them using the app, as well as what kinds of data the app would be collecting. At no point does the app capture any personally identifiable information from a participant.

Alongside this, any data taken from participants was ensured to be compliant under the Data Protection Act 1998. When ensuring the protection of user data, all information was kept in a local folder on the author's PC at his home, as well as on an external USB.

5.2 Social Issues

The projects primary subject matter involved addressing the issue of addiction pertaining to social media and smartphones. When discussing the topic of addiction, it's important that the issue is addressed in a non-stigmatizing manner and with respect to those who may be dealing with such issues . Throughout the report, the issue of addiction has been discussed in a way that doesn't stigmatize those who may be dealing with, or have dealt with such issues. As well as this, at no point within the report does the author claim the PRI to be a treatment for the issue of SA & SMA, rather, as a teaching opportunity for someone to learn about the issue.

5.2.1 Sustainable Development Goals

This project has aligned itself with Sustainable Development Goals 3 and 4. Below are the definitions provided by the United Nations:

- Ensure healthy lives and promote well-being for all at all ages .
- Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all .

Target 3.4 (Reduce Mortality From Non-communicable Diseases And Promote Mental Health) for SDG 3 states that by 2030, it should:

...reduce by one third premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being .

The project's intention from the very beginning was to encourage healthier smartphone and social media usage. It attempts to achieve this by encouraging the user to partake in activities from outside their smartphone and social media and set goals for themselves that they can achieve away from their smartphone and social media.

In relation to goal 4, a key component of the app is making the user aware of their habits, and what they can do to lower their smartphone and social media usage should it be at a high level. This is also help through the use of the WDEP system introduced by Reality Therapy, whose system intrinsically includes learning opportunities through self-evaluation. This teaching aspect was also a component that was intended to be included following the literature review.

Through the promotion of better smartphone habits, and through provided opportunities for the user to learn about their smartphone habits, the author believes the project correctly aligns with these two goals.

5.3 Ethical Issues

Arguably, the largest issue the project had to consider during its lifespan were Ethical Issues. As previously mentioned when discussing the project's primary objectives in section 1.2.2, for conduction the evaluation, human participants were required to carry out said evaluation as intended. This aspect by itself already constitutes as an ethical issue, given the use of human participants for a project of this nature raises concerns regarding the autonomy, well-being of participants, as well as the requirement of seeking ethical approval , but this specific use of human participants brings forth another issue. The evaluation involved collecting app usage data from

participants, as well as responses from the post-evaluation survey.

Similarly, during the requirements analysis stage of the project's execution, a survey was distributed, and 1-to-1 interviews were conducted. Given the use of software tools to carry out these parts of the project, this raises issues with data protection, informed consent, and the privacy of participants . Dealing with such issues required taking different precautions at each step. When ensuring the privacy of participants at any stage, identifying information was either redacted, anonymised, or not collected at all.

When conducting user interviews and the evaluation, participants were given a consent form to sign before taking part in anything related to the project, and before taking part in any survey, participants were asked to provide informed consent to take part and were given as much information as possible, something which was noted as an issue by .

To conduct any part of the project where human participants are required, ethical approval via an ethics form is required to be submitted prior to conducting any part of it. All the details written about previously were mentioned within the ethics form, alongside reasonable justification, and ethical approval was granted.

The code used for the PRI sources a small sample of the code from an external sample project as a basis for loading, and showing the usage for a user. This code has been attributed to the respective author in both the source code for the PRI, as well as the report in section 3.5.1, and within Appendix I.

5.4 Professional Issues

Following the completion of this project, there are a few aspects that have been reflected on that the author believes have undertaken well, and others the author wishes would've been undertaken differently. For one, section 3.3 details the SDLC used during the project's execution, and how there was a change in SDLC midway through the project. Instead of this conducting this change midway through the project, this change could've occurred before any work was carried out, or could've been prevented altogether if deeper research was conducted at that stage.

The author believes that issues related to transparency were handled well. Information related to how the data of participants would be used, as well what their involvement would and wouldn't entail were communicated at every possible step, and participants were offered the choice of opting out of any project-related activity at any given time, without notice. Participants were told to forward any issues

or questions they may have to the author, showing a level of accountability and responsibility taken by the author.

6 Conclusions and Further Work

6.1 Summary

Over the past fifteen years, smartphones and social media have been quickly integrated into wider society. Both of them serve as an important component of everyday life for a large amount of people and have helped connect people with each other from around the globe on a never-before-seen scale. With this however, the issue of individuals developing an overdependence on such technology has become a prominent talking point when discussing these technologies, and subsequently, studies have been carried out analysing the issue, its causes, its effects on one's health, and potential solutions, one of which being intervention apps that attempt to help users lower their usage, and another being the use of psychological methods. With that being said, a gap of intervention apps that deal with the aforementioned issue was found. The primary aim of the project was to carry out the research statement, being:

To create a mobile application that integrates psychological intervention methods alongside usage tracking technologies to help address smartphone and social media addiction

A literature review was carried out to investigate existing applications that use psychological intervention methods when addressing related issues, how SA and SMA are contextualized within existing apps, and the issues they raise, existing psychological frameworks that can be integrated into a mobile application, and methods to track a user's app usage.

Following the findings from the literature review, an effort to create a mobile application that used usage tracking technologies and psychological intervention methods while following a "develop and test" approach began. A development lifecycle was chosen, initially being an agile methodology before switching to a hybrid methodology, an analysis of the requirements necessary for the app to function as intended was undertaken, as well as picking a psychological intervention and usage tracking method, those being Reality Therapy and using the android.app.usage package, respectively. Flowchart design diagrams showing the app flow for each requirement and UI wireframes that depict how each app screen should look were then created. After this, the process of coding the app based on the information created and obtained began, with close attention paid to the prior information. After each iteration, testing was performed to ensure the app worked without issue. Finally, an evaluation with multiple participants took place where they used the app over a three-week span and sent their usage data for further analysis at one week intervals,

before answering a survey at the end of the three-week period.

6.2 Results

To reiterate what was stated in section 4, based off of the hypothesis, the intention of the evaluation was to find out if:

1. The mobile phone usage from a participant decreased when comparing their usage from the start of the evaluation to the end.
2. If there was a decrease in usage, was the PRI actually a contributing factor to it?

6.2.1 Was there a decrease in usage?

The quantitative data gathered and presented in section 4.1 reveal that only two of the four participants usage decreased overall when comparing their usage at the start to the end of the evaluation period, with this result staying consistent when only social media apps are included in the dataset. Therefore, this first point is proven to be plausible, or somewhat true.

6.2.2 Was the PRI a factor in a reduction in usage?

The qualitative data gathered and presented in section 4.2 reveal that participants who experienced a usage drop at some point during the evaluation period positively attributed the app to be a significant factor in, and carry the opinion that the app helped them achieve their set goal. Therefore, this second point is proven to be true beyond a reasonable doubt.

6.2.3 Conclusion

To conclude, the PRI and project as a whole successfully satisfies the research statement and project hypothesis, however, does so to an extent. The project does indeed create a mobile app that uses psychological intervention methods with usage tracking technologies to address the issue of SA and SMA, and has also shown to reduce the smartphone usage in young adults, however, from the results of the evaluation, the PRI only reduced the smartphone and social media usage in two of the four participants, though this is juxtaposed from the opinions from the participants themselves, who believe that the app played a significant role in reducing their smartphone and social media usage.

6.3 Limitations & Project Critique

Despite proving the research statement and project hypothesis to be true, the project is limited in some regards. A limitation of the project was the use of only one psychological intervention method. The project only uses a single concept from a single psychological framework, that being the WDEP system from Reality Therapy. The decision to use a single intervention method from a single framework was made to limit scope creep and to keep development as simple as possible. Within the intro and literature review, CBT is referenced as a psychological framework that has seen use and success in apps similar to the one made for the project, so, integrating frameworks such as ones seen in CBT would be an area of improvement.

Although not a issue for the current iteration of the project, from September of 2026 onwards, apps to be installed on Android devices in certain regions will have to be from verified Android developers . This means that future research that involves evaluating an app on a real device with human participants will require a researcher to be an Android verified developer, or go through the process of side-loading the app .

The project was limited during the evaluation phase, due to the number of participants involved. While the intention was to gather ten participants for the evaluation, only four participants were gathered. The reasoning behind this lacklustre turnout may be due to poor recruiting efforts when it came to gathering participants, as recruiting efforts were limited to a small group of people. Should a project of this calibre be attempted again, promoting the evaluation to a wider group of individuals, perhaps in physical areas such as university campuses may prove to be an effective method of gathering participants.

6.4 Further Work

When it comes to other work this project could potentially influence, investigating what psychological framework between CBT, RT, or another psychological framework in an app of this nature, and which one proves to be more beneficial from a quantitative and qualitative standpoint could an area that could use further coverage, and this could influence further use of a framework, establishing it as a standard when addressing the issue of SA and SMA.

On the contrary, and as alluded to in the previous section, future work could benefit from integrating more than one psychological intervention method, such as CBT, and its various methods mentioned in the literature review such as behavioural activation, cognitive restructuring, and problem solving. From a technical standpoint,

using app restriction functionality with usage tracking alongside either, or both psychological frameworks could shed further light on this topic area.

Expanding the target audience to include other age groups such as adolescents or older individuals could prove beneficial when evaluating apps of this nature in future studies, as it could provide deeper insights into the smartphone and social media usage habits of those groups.